

■ 1.7.13 Advanced Topic: Combinatorial Operations

You can use lists to set up many kinds of combinatorial calculations. Here are a few examples.

<code>Permutations[list]</code>	give all possible orderings of <i>list</i>
<code>Outer[List, list₁, list₂, ...]</code>	give lists of elements in the <i>list_i</i> combined in all possible ways
<code>OrderedQ[list]</code>	give <code>True</code> if the elements of <i>list</i> are in order
<code>Signature[list]</code>	give the signature of the permutation needed to put <i>list</i> into standard order

Some combinatorial operations on lists.

This gives the $3! = 6$ possible permutations of three elements.

```
In[1]:= Permutations[{a,b,c}]
Out[1]= {{a, b, c}, {a, c, b}, {b, a, c}, {b, c, a},
          {c, a, b}, {c, b, a}}
```

This combines the list elements in all possible ways. This operation is analogous to a mathematical “outer product” (see Section 3.7.12).

```
In[2]:= Outer[List, {a, b}, {c, d}]
Out[2]= {{a, c}, {a, d}, {b, c}, {b, d}}
```